

THE ENGINEER'S CODE 2026 — TECHNICAL MODULES

OVERVIEW

SENTEC, NED University of Engineering and Technology

# 1 RAG CHATBOT BUILDER	
Statement	You have been hired as an AI engineer by a growing edtech startup. The support team is overwhelmed by repetitive product questions. The client needs a chatbot deployed within 48 hours that answers strictly from the provided manual, never invents features, and stays composed with difficult users.
What To Do	Build a Retrieval-Augmented Generation chatbot from a provided PDF that answers accurately and stays polite under a live adversarial roleplay.
Description	Teams receive a PDF (handbook or manual) and must build a full RAG pipeline: document chunking, embedding, vector storage, retrieval, and grounded generation with a polished system prompt. Live judging includes a roleplay attack by a judge.
Tools	LangChain or LlamaIndex; ChromaDB or FAISS; OpenAI text-embedding-3-small or Sentence-Transformers; Anthropic Claude API or OpenAI GPT API; PyPDF2 or pdfplumber
Technical Part	Chunking strategy, embedding quality, retrieval relevance, hallucination control
Creative Part	Persona and tone design, graceful handling of out-of-scope questions, de-escalation under attack
Reference Material	Organizer-provided PDF
How Teams Build It	Pipeline built over two days, tested against the source document, then evaluated live
Rubrics	Retrieval accuracy and grounding; tone and persona design; composure under live attack
Feasibility	Moderate
Difficulty	Moderate to Hard